

Safety Data Sheet in accordance with
GB/T 16483 and GB/T 17519
Synergen B 3155



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	First compiled on: 2015/01/15
Substance key: 000000576153	Revision Date: 2021/02/23
Version : 1 - 2 / CHI	Date of printing : 2023/02/15

1. PRODUCT AND COMPANY IDENTIFICATION

Trade name

Synergen B 3155

Material number: 285454

Substance name: N,N-Dihydroxyethylcyclohexylamine water solution

Use of the Substance/Mixture

Industry sector : Industrial Performance Chemicals

Type of use : Techno-chemical industry.

Identification of the company

Clariant Specialty Chemicals
(Zhenjiang) Co. Ltd.
39,Linjiang Xi Road,New District,Zhenjiang City,Jiangsu Pro.
212132 Zhenjiang City
Telephone : +86 511 8595 6637

Information about the substance/preparation

BU Care Chemicals
Product Stewardship
E-mail address: SDS.China@clariant.com

Emergency telephone number: +86 532-83889090
(For product with GHS classification (Qingdao NRCC))

+86 21 37911379
(For product without GHS classification (Jinshan Shanghai))

2. HAZARDS IDENTIFICATION

Emergency Overview

Appearance	: liquid
Colour	: not specified
Odour	: not specified

Causes severe skin burns and eye damage. May cause damage to organs through prolonged or repeated exposure if swallowed. Harmful if swallowed. Harmful to aquatic life with long lasting effects.

GHS Classification

Acute toxicity (Oral)	: Category 4
Skin corrosion/irritation	: Category 1C
Serious eye damage/eye irritation	: Category 1
Specific target organ toxicity - repeated exposure (Oral)	: Category 2

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Long-term (chronic) aquatic hazard : Category 3

GHS label elements

Hazard pictograms : 

Signal word : Danger

Hazard statements : H314 Causes severe skin burns and eye damage.
H373 May cause damage to organs through prolonged or repeated exposure if swallowed.
H302 Harmful if swallowed.
H412 Harmful to aquatic life with long lasting effects.

Precautionary statements :

Prevention:

P260 Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
P264 Wash skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P301 + P310 + P330 IF SWALLOWED: Immediately call a POISON CENTER/ doctor. Rinse mouth.
P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower.
P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
P314 Get medical advice/ attention if you feel unwell.
P363 Wash contaminated clothing before reuse.

Storage:

P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

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Physical and chemical hazards

Not classified based on available information.

Health hazards

Harmful if swallowed. Causes severe skin burns and eye damage. Causes serious eye damage. May cause damage to organs through prolonged or repeated exposure if swallowed.

Environmental hazards

Harmful to aquatic life with long lasting effects.

Other hazards which do not result in classification

According to the present state of knowledge provided this product is handled correctly, there is no danger to humans or the environment

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
2,2'-(Cyclohexylimino)bisethanol	4500-29-2	>= 20 -<= 30

4. FIRST AID MEASURES

General advice : Get medical advice/ attention if you feel unwell.

If inhaled : If inhaled, remove to fresh air.

In case of skin contact : In case of contact, immediately flush skin with plenty of water.

In case of eye contact : Rinse the affected eye with plenty of water, at the same time keep the unaffected eye well protected.

If swallowed : If swallowed do not induce vomiting, seek medical advice and show safety datasheet or label

Most important symptoms and effects, both acute and delayed : No symptoms known currently.
No hazards known at this time.

Notes to physician : Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media : Water spray jet
Dry powder
Carbon dioxide (CO2)

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	Alcohol-resistant foam
Unsuitable extinguishing media	: High volume water jet
Specific hazards during firefighting	: In case of fires, hazardous combustion gases are formed: Carbon monoxide (CO) Carbon dioxide (CO ₂) Nitrogen oxides (NO _x)
Specific extinguishing methods	: Evacuate endangered area, seal off area. Prevent further leakage or spillage. Wear suitable protective equipment.
Special protective equipment for firefighters	: Self-contained breathing apparatus

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Wear suitable protective equipment.
Environmental precautions	: The product should not be allowed to enter drains, water courses or the soil.
Methods and materials for containment and cleaning up	: Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Treat recovered material as described in the section "Disposal considerations".
Prevention of secondary hazards	: Do not let the liquid drain into rivers, ponds or sewer systems.

7. HANDLING AND STORAGE

Handling

Advice on protection against fire and explosion	: Observe the general rules of industrial fire protection
Advice on safe handling	: When used and handled appropriately no special measures are needed
Avoidance of contact	: not known

Storage

Conditions for safe storage	: No special measures necessary.
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Further information on storage conditions	: Keep containers tightly closed in a cool, well-ventilated place. Handle and open container with care. Keep away from heat. Do not freeze.
Materials to avoid	: When used and handled as intended, none. Do not store or transport together with foodstuffs
Further information on storage stability	: Shelf life is limited; see product information leaflet

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Contains no substances with occupational exposure limit values.

Personal protective equipment

Respiratory protection : Protection necessary if aerosols or vapors should develop.
mask, comb.gas/particle filter

Eye/face protection : Safety glasses

Skin and body protection : working clothes

Hand protection

Remarks : Chemical resistant gloves Minimum breakthrough time (glove): not determined Minimum thickness (glove): not determined Take note of the information given by the producer concerning permeability and break through times, and of special workplace conditions (mechanical strain, duration of contact).

Protective measures : Observe the usual precautions for handling chemicals.

Hygiene measures : Wash hands before breaks and at the end of workday.
Use protective skin cream before handling the product.
Take off immediately all contaminated clothing and wash it before reuse.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid

Colour : not specified

Odour : not specified

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Odour Threshold	: cannot be determined
pH	: 10.6 - 11.2 (20 °C) Concentration: 10 g/l Method: DIN EN 1262
Melting point	: < -10 °C Method: OECD Test Guideline 102
Boiling point	: approx. > 100 °C Method: DSC
Flash point	: 127.5 °C Method: DIN EN 22719 / ISO 2719 (closed cup)
Evaporation rate	: not tested.
Self-ignition	: Not applicable
Burning number	: Not applicable
Upper explosion limit / upper flammability limit	: not tested.
Lower explosion limit / Lower flammability limit	: not tested.
Vapour pressure	: < 0.0001 hPa (20 °C) Method: OECD Test Guideline 104
Relative vapour density	: not tested.
Relative density	: not tested.
Density	: 1.02 g/cm ³ (20 °C)
Solubility(ies)	
Water solubility	: miscible (20 °C) Method: OECD Test Guideline 105
Solubility in other solvents	: not tested.
Partition coefficient: n-octanol/water	: log Pow: < 1 Method: Calculated from the solubilities
Auto-ignition temperature	: not tested.
Decomposition temperature	: 350 °C Heating rate: 3 K/min Method: DSC

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Viscosity	
Viscosity, dynamic	: not tested.
Viscosity, kinematic	: not tested.
Flow time	: not tested.
Explosive properties	: no data available
Oxidizing properties	: not tested.
Minimum ignition energy	: not tested.
Particle size	: Not applicable

10. STABILITY AND REACTIVITY

Reactivity	: See section 10.3. "Possibility of hazardous reactions"
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use. Stable
Conditions to avoid	: None known.
Incompatible materials	: not known
Hazardous decomposition products	: When handled and stored appropriately, no dangerous decomposition products are known

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Product:

Acute oral toxicity	: LD50 (Rat): approx. 8.000 mg/kg Method: OECD Test Guideline 423
Acute inhalation toxicity	: Remarks: not tested.
Acute dermal toxicity	: Remarks: not tested.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Acute oral toxicity	: LD50 (Rat, female): $\geq 2,000$ mg/kg Method: OECD Test Guideline 423
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GLP: yes

Acute inhalation toxicity : Remarks: not required

Acute dermal toxicity : Remarks: not required

Skin corrosion/irritation

Product:

Method : OECD Test Guideline 435
Result : Corrosive

Components:

2,2'-(Cyclohexylimino)bisethanol:

Species : reconstructed human epidermis (RhE)
Exposure time : 3 - 60 min
Method : OECD Test Guideline 431
Result : Corrosive after 1 to 4 hours of exposure
GLP : yes

Serious eye damage/eye irritation

Product:

Result : Corrosive
Method : OECD Test Guideline 437

Components:

2,2'-(Cyclohexylimino)bisethanol:

Species : Bovine cornea
Result : Risk of serious damage to eyes.
Exposure time : 4 h
Method : OECD Test Guideline 437
GLP : yes

Respiratory or skin sensitisation

Product:

Remarks : not tested.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Test Type : Maximisation Test
Exposure routes : Dermal
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Not a skin sensitizer.
GLP : yes

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Assessment : Harmful if swallowed., Causes severe skin burns and eye damage.

Germ cell mutagenicity

Product:

Germ cell mutagenicity - Assessment : No information available.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Genotoxicity in vitro : Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster lung cells
Concentration: 250 - 5000 µg/ml
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative
GLP: yes

Test Type: Ames test
Test system: Salmonella typhimurium
Concentration: 31,6 - 5000 µg/plate
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

Test Type: In vitro mammalian cell gene mutation test
Test system: Chinese hamster lung cells
Concentration: 70 - 5000 µg/ml
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes

Germ cell mutagenicity - Assessment : In vitro tests did not show mutagenic effects

Carcinogenicity

Product:

Carcinogenicity - Assessment : No information available.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Carcinogenicity - : No information available.

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Assessment

Reproductive toxicity

Product:

Reproductive toxicity - Assessment : No information available.

No information available.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Effects on fertility : Test Type: Fertility/early embryonic development
Species: Rat, male and female
Strain: wistar
Application Route: oral (gavage)
Dose: 0 - 12 - 30 - 75 mg/kg
General Toxicity - Parent: NOAEL: >= 75 mg/kg body weight
General Toxicity F1: NOAEL: >= 75 mg/kg body weight
Method: OECD Test Guideline 421
GLP: yes

Reproductive toxicity - Assessment : No evidence of adverse effects on sexual function and fertility, or on development, based on animal experiments.

STOT - single exposure

Product:

Remarks : not available

Components:

2,2'-(Cyclohexylimino)bisethanol:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure

Product:

Remarks : not available

Components:

2,2'-(Cyclohexylimino)bisethanol:

Exposure routes : Oral
Target Organs : Gastrointestinal tract
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

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Repeated dose toxicity

Product:

Remarks : not tested.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Species : Rat, male and female
NOAEL : 75 mg/kg bw/day
Application Route : oral (gavage)
Exposure time : 28 d
Number of exposures : once per day
Dose : 12 - 30 - 75 mg/kg
Control Group : yes
Method : OECD Test Guideline 407
GLP : yes

Repeated dose toxicity - Assessment : Harmful if swallowed., Causes severe skin burns and eye damage.

Aspiration toxicity

Product:

no data available

Components:

2,2'-(Cyclohexylimino)bisethanol:

No aspiration toxicity classification

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to fish :
Remarks: not tested.

Toxicity to daphnia and other aquatic invertebrates : EC50: 652 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants :
Remarks: not tested.

Toxicity to fish (Chronic toxicity) :
Remarks: not tested.

Toxicity to daphnia and other :
Remarks: not tested.

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aquatic invertebrates
(Chronic toxicity)

Toxicity to microorganisms : Remarks: not tested.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): > 100 mg/l
End point: mortality
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: OECD Test Guideline 203
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 140 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: yes

Toxicity to algae/aquatic plants : ErC10 (Desmodesmus subspicatus (green algae)): 8.1 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

ErC50 (Desmodesmus subspicatus (green algae)): 10.8 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

Toxicity to fish (Chronic toxicity) : NOEC (Danio rerio (zebra fish)): 100 mg/l
Exposure time: 8 d
Test Type: semi-static test
Analytical monitoring: yes
Method: OECD Test Guideline 212
GLP: yes

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 13 mg/l
End point: Reproduction rate
Exposure time: 21 d
Test Type: semi-static test

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Analytical monitoring: yes
Method: OECD Test Guideline 211
GLP: yes

Toxicity to microorganisms : EC50 (activated sludge): > 860 mg/l
End point: Bacteria toxicity (respiration inhibition)
Exposure time: 3 h
Test Type: aquatic
Analytical monitoring: no
Method: OECD Test Guideline 209
GLP: yes

Ecotoxicology Assessment

Chronic aquatic toxicity : This product has no known ecotoxicological effects.

Persistence and degradability

Product:

Biodegradability : Result: Not readily biodegradable.
Biodegradation: 24 %
Exposure time: 28 d
Method: OECD Test Guideline 301B

Physico-chemical : Remarks: not available
removability

Components:

2,2'-(Cyclohexylimino)bisethanol:

Biodegradability : aerobic
Inoculum: activated sludge
Concentration: 10 mg/l
Carbon dioxide (CO₂)
Result: Not readily biodegradable.
Biodegradation: 24 %
Exposure time: 28 d
Method: OECD Test Guideline 301B
GLP: no

Physico-chemical : Remarks: Not readily biodegradable.
removability

Photodegradation : Test Type: air
Method: other (calculated)
GLP: no

Test Type: water
Remarks: Not applicable

Test Type: Soil

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Remarks: Not applicable

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: not tested.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Bioaccumulation : Remarks: Low potential for bioaccumulation (log Pow < 3).

Partition coefficient: n-
octanol/water : log Pow: 0 (20 °C)
Method: Other
GLP: no

Mobility in soil

Product:

Distribution among
environmental compartments : Remarks: not tested.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Distribution among
environmental compartments : Remarks: Not expected to adsorb on soil.

Other adverse effects

Product:

Environmental fate and
pathways : Remarks: not available

Results of PBT and vPvB
assessment : Remarks: no data available

Additional ecological
information : The product should not be allowed to enter drains, water
courses or the soil.

Components:

2,2'-(Cyclohexylimino)bisethanol:

Environmental fate and
pathways : not available

Results of PBT and vPvB
assessment : The substance is not identified as a PBT or as a vPvB
substance.

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13. DISPOSAL CONSIDERATIONS

Disposal methods

- Waste from residues : Product should be taken to a suitable and authorized waste disposal site in accordance with relevant regulations and if necessary after consultation with the waste disposal operator and/or the competent Authorities
- Contaminated packaging : Packaging that cannot be cleaned should be disposed of as product waste

14. TRANSPORT INFORMATION

IATA

Proper shipping name: Amines, liquid, corrosive, n.o.s.
Class: 8
Packing group: III
UN/ID number: UN 2735
Primary risk: 8
Remarks: Shipment permitted
Hazard inducer(s): CYCLOHEXYLAMINE

IMDG

Proper shipping name: Amines, liquid, corrosive, n.o.s.
Class: 8
Packing group: III
UN no. UN 2735
Primary risk: 8
Remarks: Shipment permitted
Hazard inducer(s): CYCLOHEXYLAMINE
EmS : F-A S-B

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TDG_CN

Proper shipping name: Amines, liquid, corrosive, n.o.s.
Class: 8
Packing group: III
UN no.: UN 2735
Primary risk: 8
Remarks: Shipment permitted
Hazard inducer(s): CYCLOHEXYLAMINE

RID

Proper shipping name: Amines, liquid, corrosive, n.o.s.
Class: 8
Packing group: III
UN no.: UN 2735
Primary risk: 8
Hazard no. : 80
Remarks: Shipment permitted
Hazard inducer(s): CYCLOHEXYLAMINE

15. REGULATORY INFORMATION

National regulatory information

GB 6944-2012: Classification and code of dangerous goods.

GB 12268-2012: List of dangerous goods.

GB 190-2009: Packing symbol of dangerous goods.

GB 30000.2-29 – 2013 , Rules for classification and labelling of chemicals.

Law of the People's Republic of China on Prevention and Control of Occupational Diseases (Amended in 2011).

GB13690: General rule for classification and hazard communication of chemicals

16. OTHER INFORMATION

Further information

Other information : Observe national and local legal requirements

Date format : yyyy/mm/dd

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized

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System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

Disclaimer

This information corresponds to the present state of our knowledge and is intended as a general description of our products and their possible applications. Clariant makes no warranties, express or implied, as to the information's accuracy, adequacy, sufficiency or freedom from defect and assumes no liability in connection with any use of this information. Any user of this product is responsible for determining the suitability of Clariant's products for its particular application. Nothing included in this information waives any of Clariant's General Terms and Conditions of Sale, which control unless it agrees otherwise in writing. Any existing intellectual/industrial property rights must be observed. Due to possible changes in our products and applicable national and international regulations and laws, the status of our products could change. Material Safety Data Sheets providing safety precautions, that should be observed when handling or storing Clariant products, are available upon request and are provided in compliance with applicable law. You should obtain and review the applicable Material Safety Data Sheet information before handling any of these products. For additional information, please contact Clariant.

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